

# Genomas: Artificial Intelligence Platform for Precision Oncology Based on Genomic Sequencing

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## Introduction

Cancer is the second leading cause of death in Chile, with over 28,000 deaths annually and more than 53,000 new cases projected by 2025. Although precision oncology has shown significant benefits — improving response rates by 20–40% and reducing toxicities — its implementation in Chile remains limited.

The main challenge lies in translating complex genomic data from next-generation sequencing (NGS) into clinically interpretable, standardized information. Current workflows produce lengthy reports in English, with limited prioritization or clinical usability, delaying decision-making in oncology.

This reflects a critical gap between bioinformatics capacity and clinical integration, characterized by lack of automation, standardization, and interoperability.

## Objective

GENOMAS aims to develop an AI-driven platform for precision oncology, automating the end-to-end process from raw sequencing data to standardized, actionable clinical reports in Spanish.

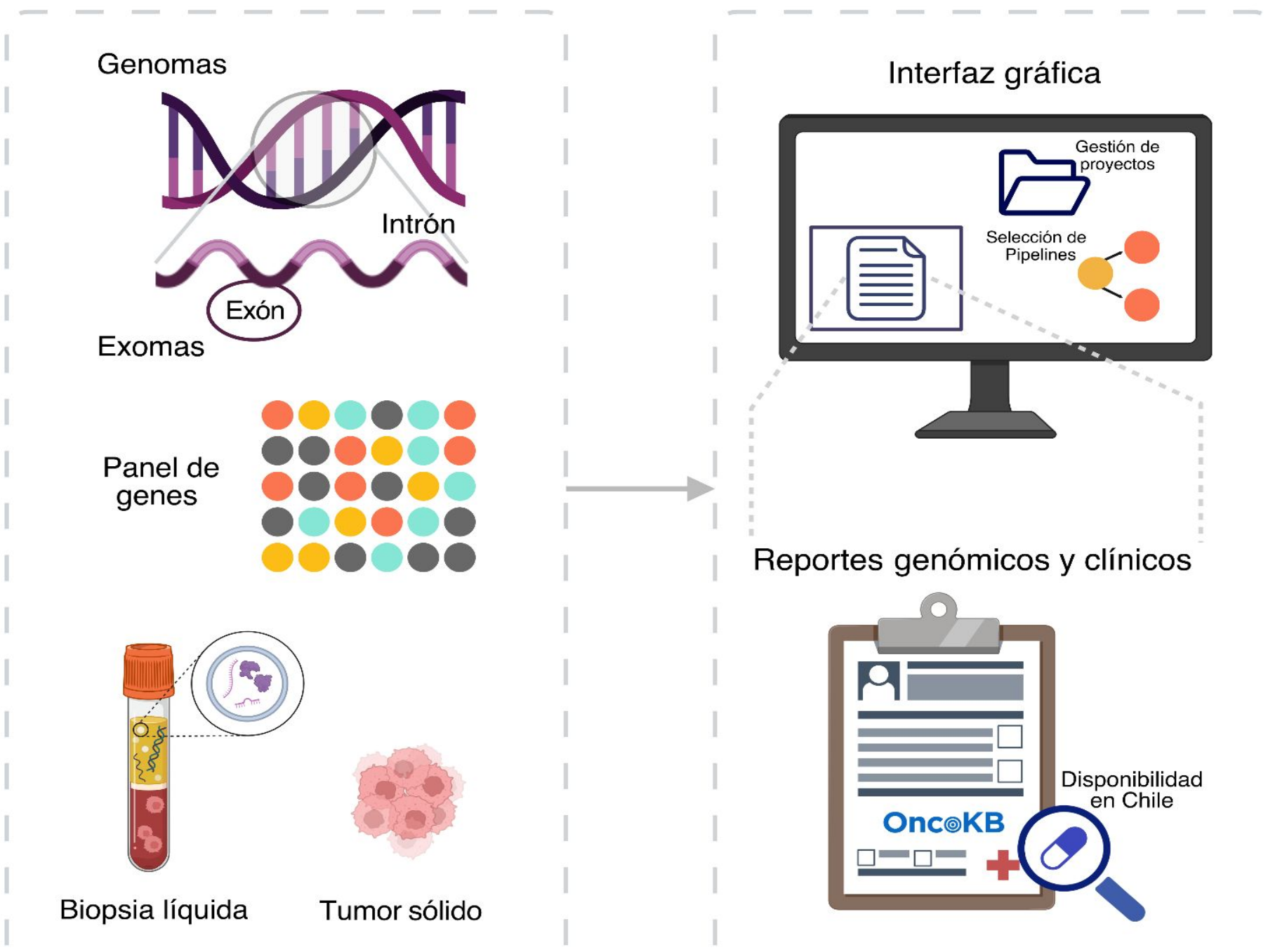
The platform bridges the “data-to-decision” gap by combining bioinformatics, AI, and clinical expertise to provide interpretable reports aligned with international guidelines and adapted to the Chilean healthcare system.

## Methodology

GENOMAS integrates a modular workflow combining genomics, bioinformatics, and artificial intelligence:

- 1. Genomic Input Layer** – Processes data from solid tumors and liquid biopsies (WGS/WES/panel-based), supporting FASTQ/BAM/VCF formats.
- 2. Computational Engine** – Automates variant calling, annotation, and interpretation pipelines with full traceability and parameterization.
- 3. AI Module** – Employs machine learning models to predict tumor type and prioritize actionable findings.
- 4. Clinical Reporting Layer** – Generates standardized, Spanish-language clinical reports, ranked by therapeutic evidence (OncoKB, ClinVar).

The system adheres to international standards for data security and interoperability (HL7 FHIR, ISO 27001/13485, Chilean Law 19.628) and will be deployed as a secure, cloud-based web platform integrable with hospital systems.

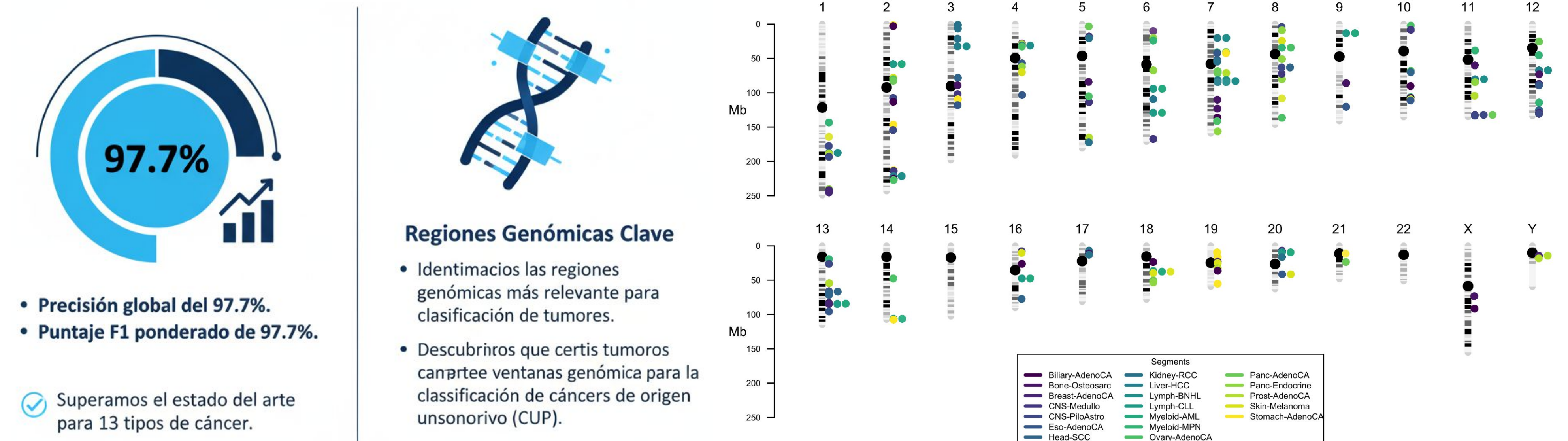


## Preliminary Results

Recent research has developed machine learning models capable of classifying cancers of unknown primary origin with >75% F1-score accuracy using mutational signatures from the PCAWG dataset. These efforts established standardized NGS analysis pipelines and validated models for tumor origin prediction, forming the technical foundation of GENOMAS.

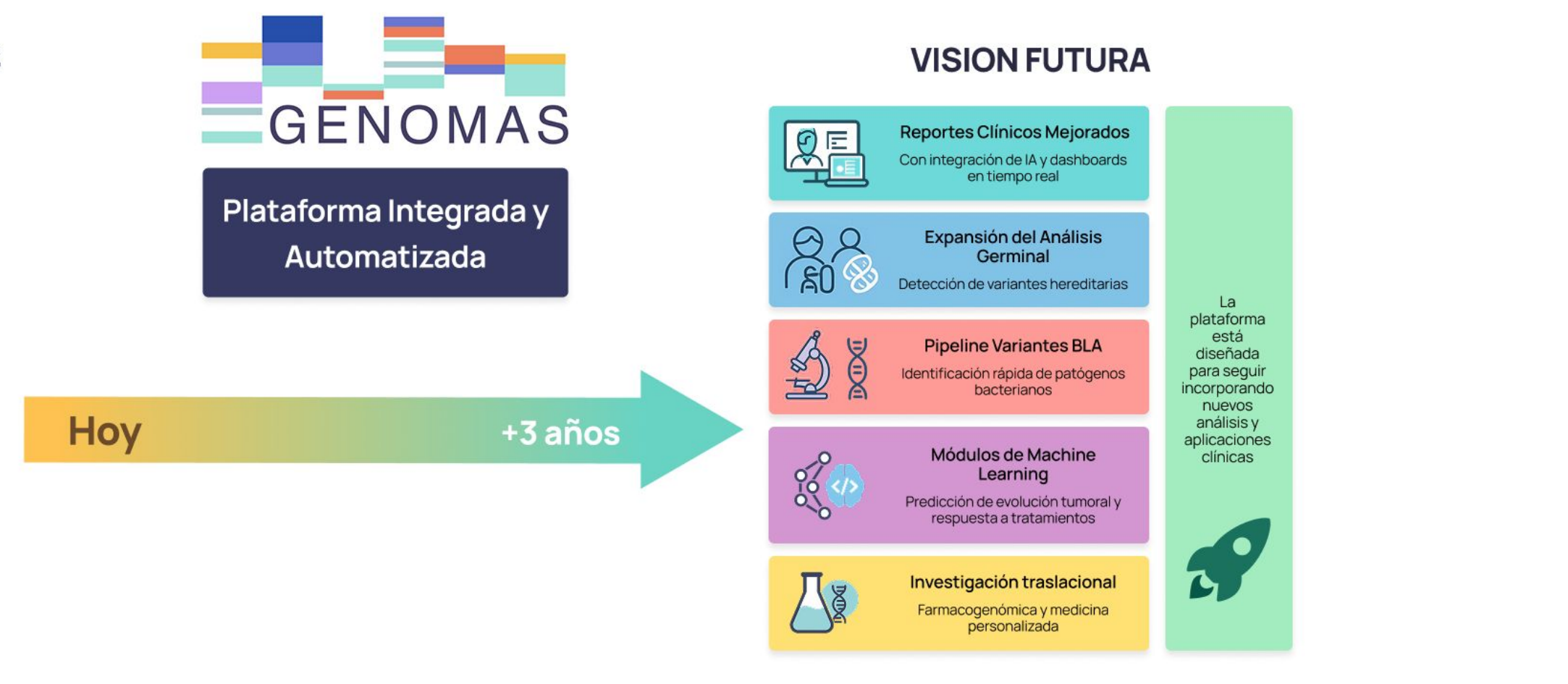
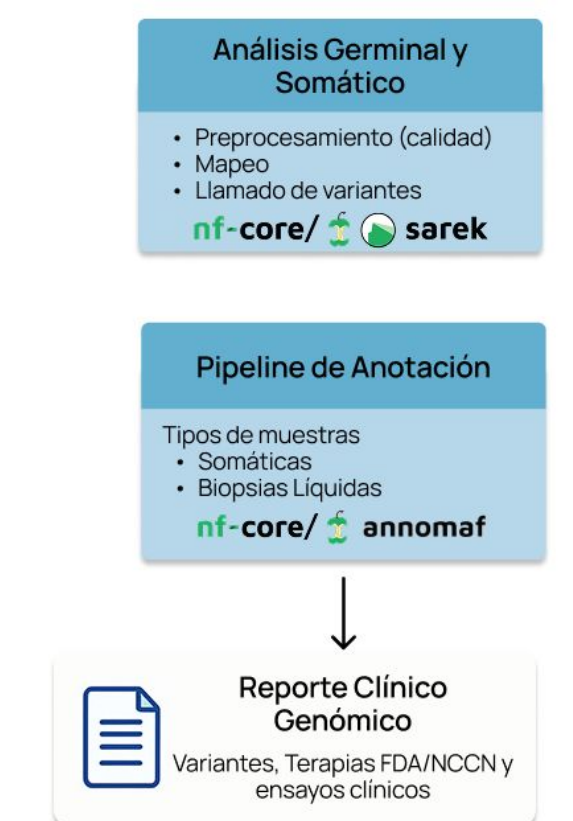
The team has published multiple peer-reviewed studies in high-impact journals, demonstrating strong expertise in AI and clinical genomics integration.

### 1. Machine Learning Module

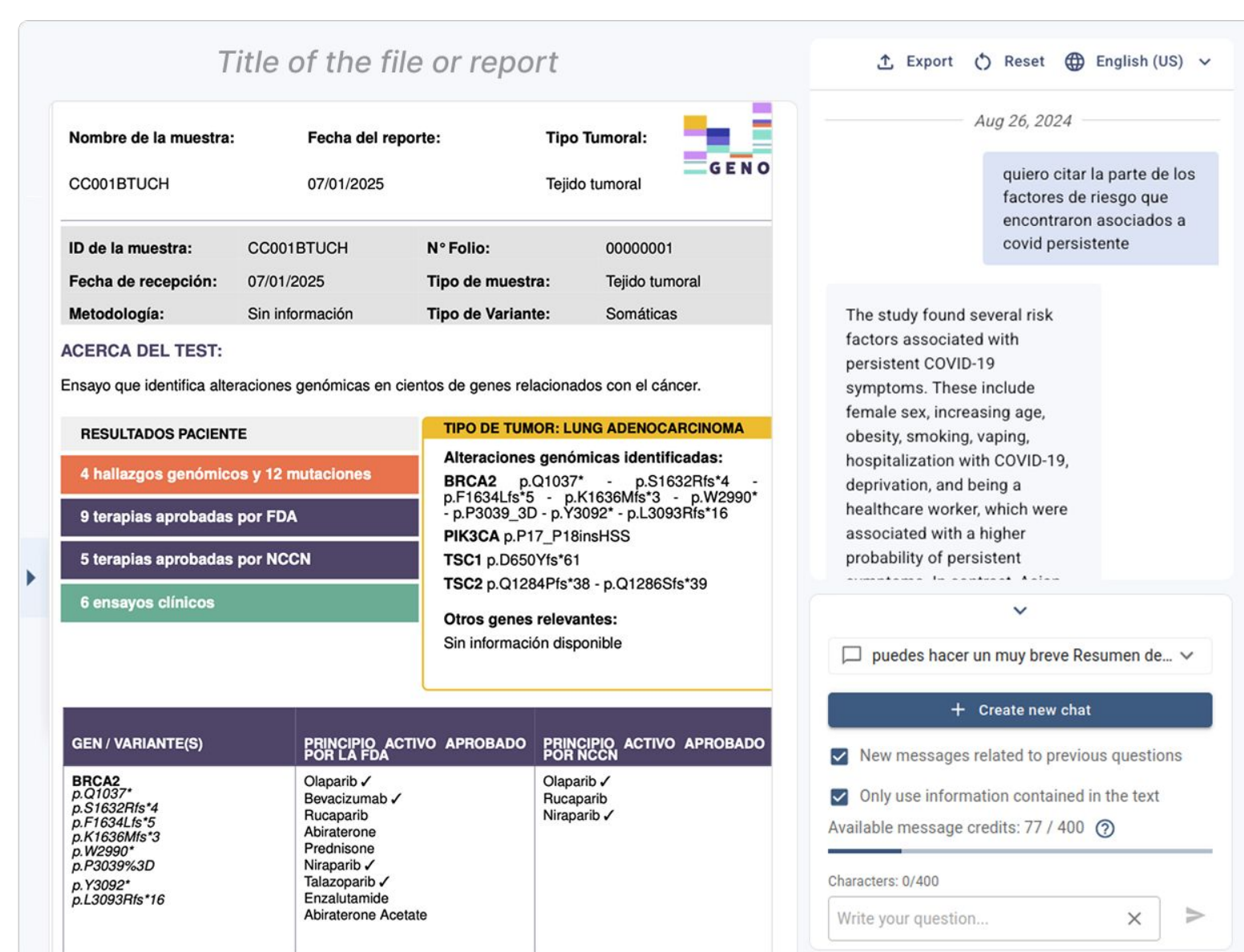


### 2. Bioinformatic module

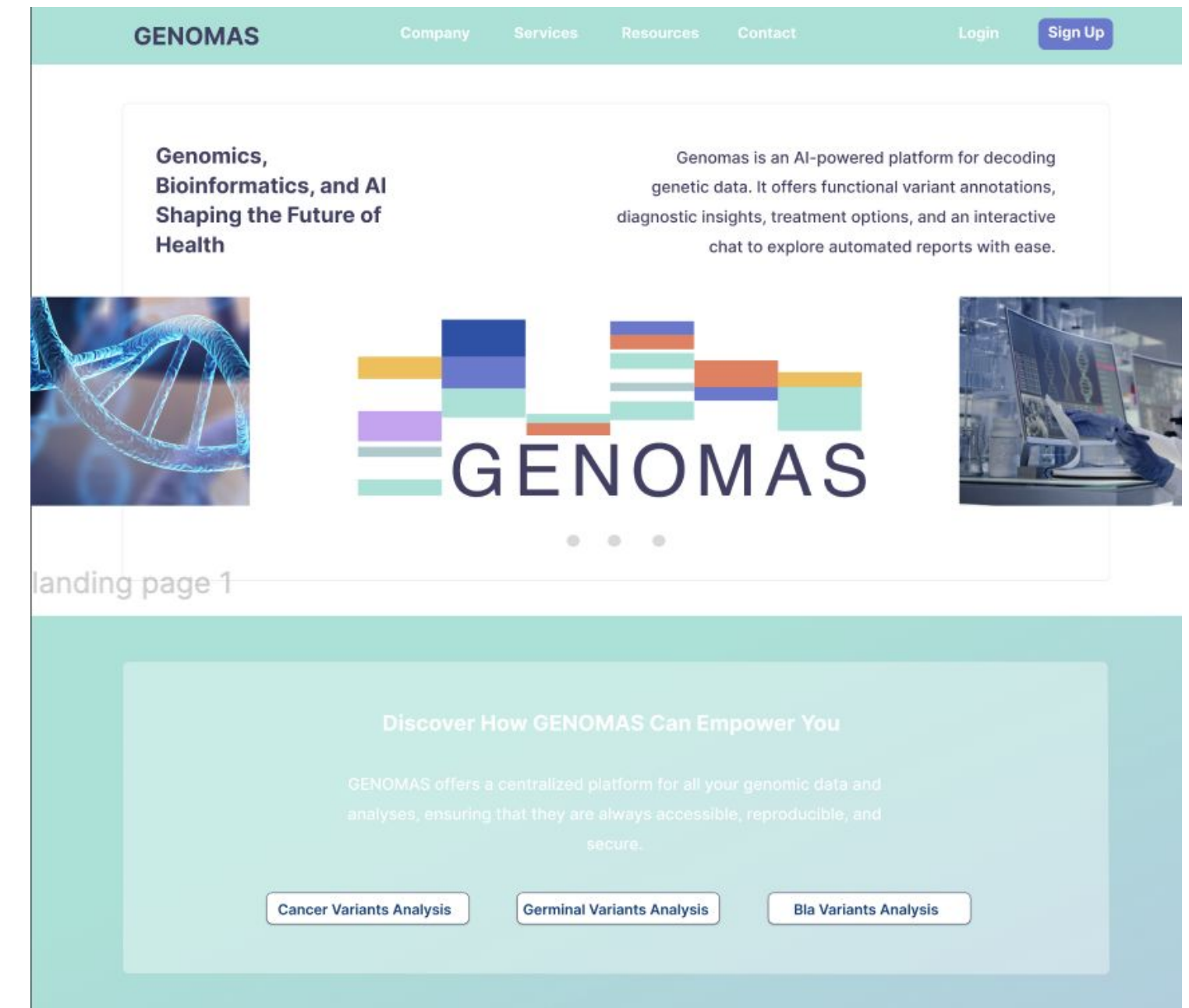
#### CAPACIDADES ACTUALES



### 3. Annotation and report module



### 4. Graphical implementation



## Conclusions and Impact

GENOMAS represents a technological milestone for precision oncology in Chile, offering an interoperable, Spanish-language solution for automated genomic interpretation.

The platform aims to reduce turnaround times from 7–14 days to 2–4 hours, achieve ≥95% concordance with international variant databases, and cut interpretation costs by over 30%.

By enabling equitable access to advanced molecular interpretation across regions, GENOMAS promotes clinical consistency, data traceability, and healthcare equity.

Future stages include multicenter clinical validation, regulatory certification, and scaling to Latin America, positioning Chile as a regional leader in precision oncology infrastructure.

## Acknowledgments

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## References

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[2] Oróstica, Karen Y., Sebastian B. Mohr, Jonas Dehning, et al. 2024. “Early Mutational Signatures and Transmissibility of SARS-CoV-2 Gamma and Lambda Variants in Chile.” *Scientific Reports* 14 (1): 16000. <https://doi.org/10.1038/s41598-024-66885-2>